

Multiple Choice (10 marks)

1. Which of the following would most likely form a fossil?
 - (a) a rock
 - (b) a crystal
 - (c) a mineral
 - (d) a bone
2. To determine how closely related organisms are, scientists consider all of the following except
 - (a) embryonic structures.
 - (b) homologous structures.
 - (c) analogous structures.
 - (d) vestigial structures.
3. About how long does it take Earth to make one revolution around the Sun?
 - (a) a day
 - (b) a week
 - (c) a month
 - (d) a year
4. An infection caused by a cut in the skin is the direct result of
 - (a) gene mutation.
 - (b) an allergic reaction.
 - (c) tissue overgrowth.
 - (d) foreign cell invasion.
5. Which task in the kitchen is an example of a physical change?
 - (a) frying an egg
 - (b) baking a cake
 - (c) melting butter
 - (d) toasting bread

True / False (10 marks)

Answer True or False

6. True or False: The balanced equation for the reaction of magnesium nitrate and potassium carbonate is $\text{Mg}(\text{NO}_3)_2 + \text{K}_2\text{CO}_3 \rightarrow \text{MgCO}_3 + 2\text{KNO}_3$.
7. True or False: Presenting information about a new species to the scientific community enhances scientific knowledge and collaboration.
8. True or False: Paper does not melt into a liquid state when heat is applied, unlike substances such as butter or candle wax.
9. True or False: Treating male insects to prevent sperm production can help reduce the total population of insects.
10. True or False: When blue light shines on a yellow banana, it appears green due to the mixing of colors.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the unique functions of plant cells that differentiate them from animal cells, focusing on specific structures and their roles in the life processes of plants.
12. Discuss the advantages of using renewable resources over non-renewable resources, and analyze how they impact the environment, economy, and future sustainability.

End of Paper